A Project Report On

Resume Analyzer & Job Recommendation System Using Machine Learning

Submitted in partial fulfillment of the requirement for the B.Sc. (D.S.) Semester-4 Mini Project Subject Final Examination.

Bachelor of Science in Data Science (B.Sc.-D. S)

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1. SYNOPSIS

This project presents an AI-based Resume Analyzer and Job Recommendation System that uses Natural Language Processing (NLP) and Machine Learning techniques to classify resumes and recommend suitable job opportunities. The system processes unstructured resume text, applies TF-IDF vectorization, and uses a Logistic Regression model to predict the most relevant job category. Based on the predicted category and resume content, a recommendation engine suggests relevant job roles by computing similarity between the resume and job descriptions. The system is implemented using Python, Scikit-learn, and FastAPI, with a user-friendly interface.

2. PREAMBLE

2.1 General Introduction

In today's competitive job market, analyzing resumes manually is inefficient. This project automates resume screening using machine learning and NLP techniques. It classifies resumes into job categories and recommends suitable job roles.

2.2 Statement of Problem

Manual resume screening is time-consuming and prone to human error. There is a need for an automated system to classify resumes and suggest relevant jobs.

2.3 Objectives and Scope

Objectives:

- Build a resume classification model
- Extract features from text
- Recommend jobs based on similarity
- Develop a web interface

Scope:

- Helps job seekers find suitable jobs
- Supports multiple domains
- Extendable to real-time systems

2.4 Module Description

1. Data Preprocessing Module
2. Feature Extraction (TF-IDF)
3. Classification Module (Logistic Regression)
4. Recommendation Module
5. Web Interface (FastAPI & HTML & CSs)
6. File handler

3. SYSTEM DESIGN AND DEVELOPMENT

3.1 Data Flow

User → Resume Input → Preprocessing → TF-IDF → ML Model → Prediction → Recommendation → Output

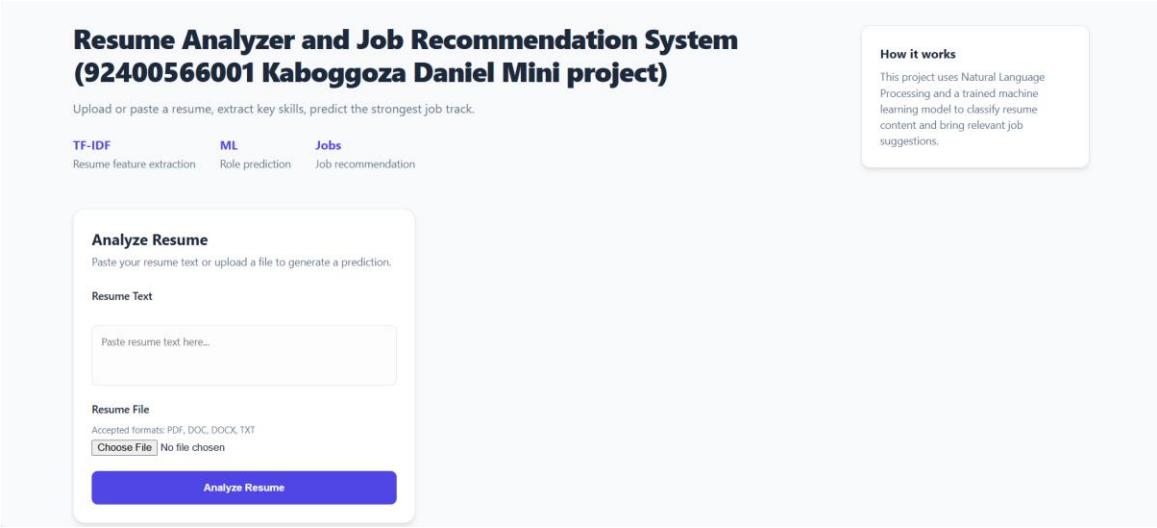
3.2 Table Structure

jobs.csv contains:

Job Title, Company, Location, Category, Description

3.3 Menu Design

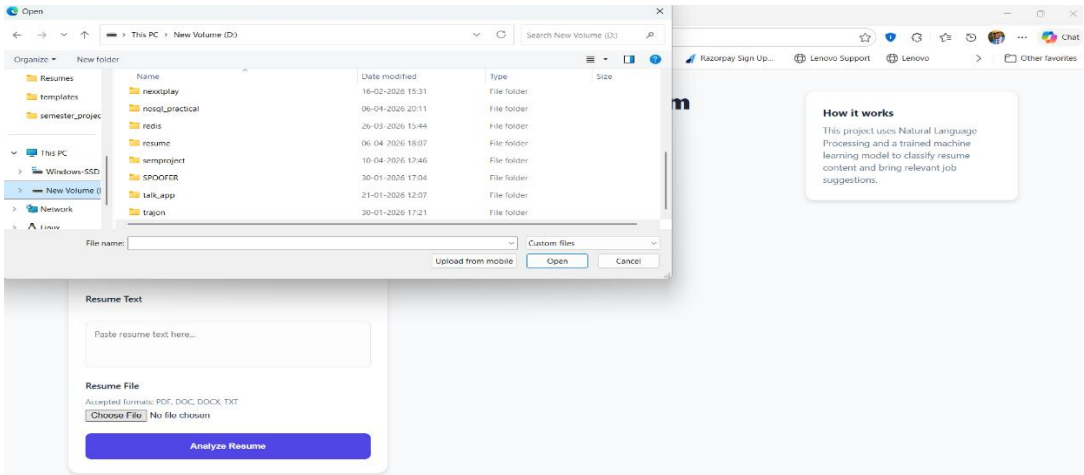
Home Page, Resume Input, Analyze Button, Results Display, Job Recommendations



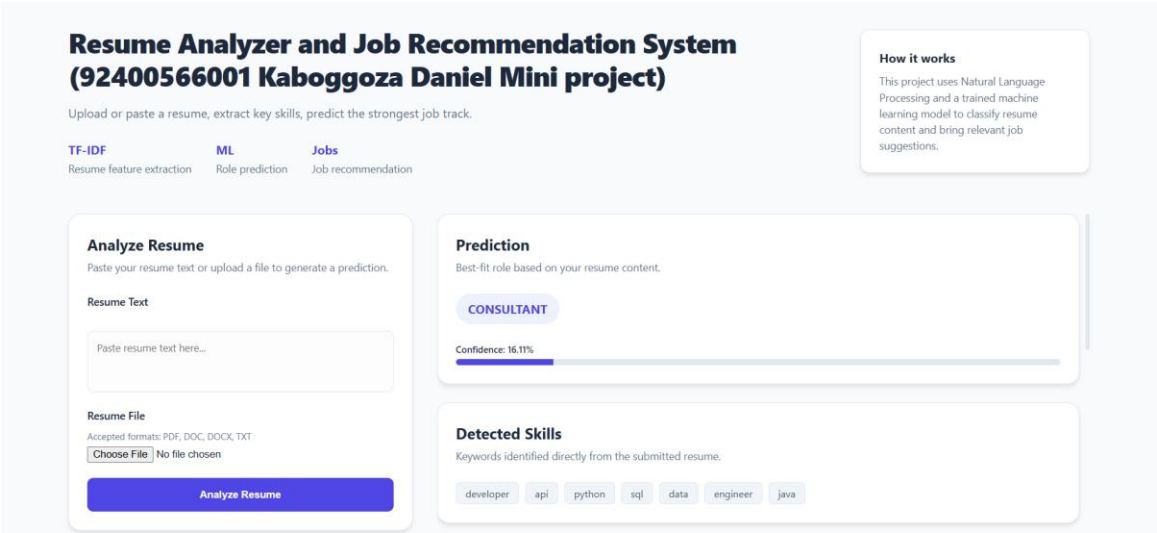
(Fig1)

3.4 Screen Design

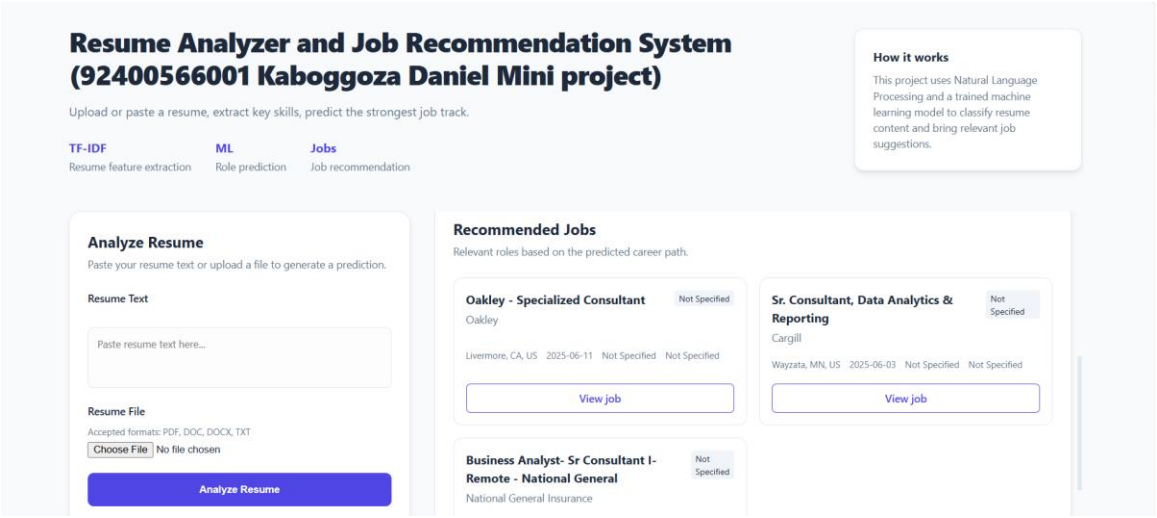
The interface allows users to input resumes and view predictions, confidence scores, and recommended jobs.



(Fig2)


Resume Analyzer and Job Recommendation System (92400566001 Kaboggoza Daniel Mini project)
Upload or paste a resume, extract key skills, predict the strongest job track.
TF-IDF
Resume feature extraction
ML
Role prediction
Jobs
Job recommendation
How it works
This project uses Natural Language Processing and a trained machine learning model to classify resume content and bring relevant job suggestions.
Analyze Resume
Paste your resume text or upload a file to generate a prediction.
Resume Text
Paste resume text here...
Resume File
Accepted formats: PDF, DOC, DOCX, TXT
Choose File No file chosen
Analyze Resume
Prediction
Best-fit role based on your resume content.
CONSULTANT
Confidence: 16.11%
Detected Skills
Keywords identified directly from the submitted resume.
developer
api
python
sql
data
engineer
java

(Fig3)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This project uses Natural Language Processing and a trained machine learning model to classify resume content and bring relevant job suggestions.
Analyze Resume
Paste your resume text or upload a file to generate a prediction.
Resume Text
Paste resume text here...
Resume File
Accepted formats: PDF, DOC, DOCX, TXT
Choose File No file chosen
Analyze Resume
Recommended Jobs
Relevant roles based on the predicted career path.
Oakley - Specialized Consultant
Not Specified
Oakley
Livermore, CA, US 2025-06-11 Not Specified Not Specified
View job
Sr. Consultant, Data Analytics & Reporting
Not Specified
Cargill
Wayzata, MN, US 2025-06-03 Not Specified Not Specified
View job
Business Analyst- Sr Consultant I- Remote - National General
Not Specified
National General Insurance

(Fig4)

4. CONCLUSION

The project demonstrates the use of ML and NLP in resume classification and job recommendation systems.

5. LEARNING DURING PROJECT WORK

Learned ML model building, NLP preprocessing, FastAPI integration, and system design.

6. BIBLIOGRAPHY

Online: Scikit-learn docs, FastAPI docs, Kaggle